

Software Engineering Project Proposal

(BDS-4B)

UniThrift

Group Members:

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1. Project Details

- **Project Title:** UniThrift: A Peer-to-Peer Academic Resource Exchange and Emergency Borrowing System
- **Project Type:** Web Application (Responsive)

2. Problem Statement

University students face a recurring financial burden caused by the high cost of academic resources such as textbooks, specialized engineering kits, and drafting tools. These items are often required for only a single semester, after which they remain unused and gather dust in dorm rooms. Meanwhile, incoming juniors are forced to purchase the same items at full retail price due to the lack of a structured resale platform. Beyond the financial aspect, students frequently encounter "equipment emergencies," such as forgetting a lab coat or a specific calculator right before a high-stakes lab or examination. Currently, there is no centralized system to facilitate quick, short-term borrowing within the campus. Relying on fragmented social media groups is often unreliable and slow, leading to avoidable academic setbacks. This project addresses the lack of a sustainable, community-driven "circular economy" that could solve these logistical and financial hurdles.

3. Problem Solution and Scope

UniThrift provides a digitized solution to the traditional "senior-to-junior" hand-off by creating a secure, verified marketplace exclusive to the university community. The platform allows students to list items for resale, donation, or short-term lending through a real-time "SOS" request system. By restricting access to verified university email domains, the solution ensures a high level of security and accountability.

The **scope** of this project includes the development of a responsive web application that manages user profiles, item catalogs, and an in-app messaging interface for coordination. It covers the logic for departmental filtering and a peer-to-peer rating system to ensure reliability. However, the scope is limited to transactions within the physical campus; the platform will facilitate

connections but will not handle physical logistics, shipping, or integrated payment gateways to avoid legal complexities. The system will primarily focus on academic-related categories like Electronics, Stationery, and Books to maintain the platform's utility as an educational support tool.

4. Basic Features

- **Item Listing Management:** A portal where users can upload high-quality photos, write detailed descriptions, and set prices or mark items as "Free".
- **SOS Emergency Borrowing Feed:** A real-time, priority-based notification system where students can post urgent requests for short-term item loans.
- **Departmental & Category Filtering:** Advanced search functionality that allows users to sort listings by specific academic departments such as CS, EE, or BBA.
- **Secure In-App Messaging:** A private chat interface that enables buyers and sellers to negotiate terms and set campus meeting spots without sharing personal phone numbers.
- **Trust and Reliability Ratings:** A peer-review mechanism where users rate each other after transactions to build a transparent reliability score for every student.
- **Personalized Activity Dashboard:** A central hub for users to track their active listings, purchase history, and items currently out on loan.
- **Automated Wishlist Alerts:** A feature allowing students to "heart" specific items and receive immediate notifications if the seller reduces the price.
- **Relational Transaction History:** A digital audit log that tracks the lifecycle of an item from listing to final sale or successful return of a borrowed tool.
- **Admin Moderation Dashboard:** A management interface for student administrators to flag inappropriate content, resolve disputes, or suspend "ghosting" users.
- **Department-Specific Trending:** A dynamic section on the home page highlighting highly requested items or popular textbooks within the user's specific department.

5. Project Plan

Iteration	Module	User Stories	Deliverables
Iteration 1 (Weeks 1-2)	User Management & Foundation	1.1: University Email Sign-up 1.2: Profile Management 1.3: Secure Authentication	Functional Authentication system and Database Schema with User Profiles.
Iteration 2 (Weeks 3-4)	Marketplace & Discovery	2.1: Item Upload/Listings 2.2: Departmental Filtering 2.3: Wishlist Implementation	Working Marketplace grid with CRUD operations for items and search functionality.
Iteration 3 (Weeks 5-6)	Logistics & Accountability	3.1: SOS Borrowing Requests	Real-time SOS feed, integrated chat system, and peer-review logic for final submission.

		3.2: In-App Messaging	
		3.3: Trust Rating System	

Name	Role	Responsibilities
Muhammad Soban Sohail	Lead & Database Architect	Designing SQL relational schemas, managing transaction integrity, and overseeing project progress.
Saad Jahangir	Full-Stack Developer	Implementing backend logic in Node.js and building the React.js frontend components.
Mahad Jawad Rana	Requirement Engineer & QA	Mapping user stories to features, drafting acceptance criteria, and conducting system testing.